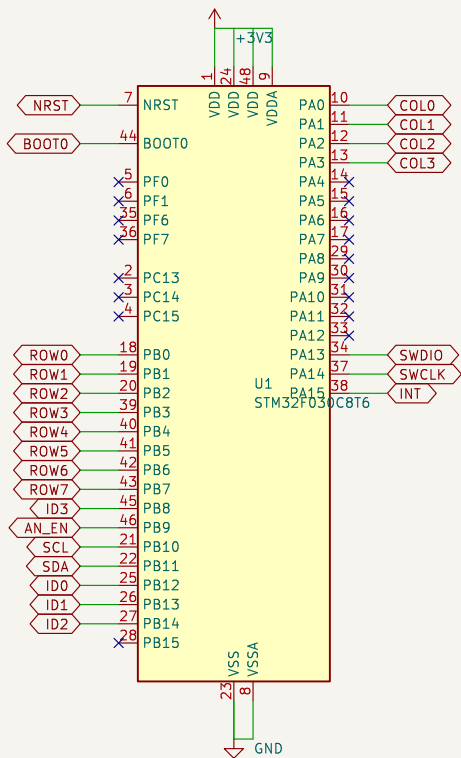
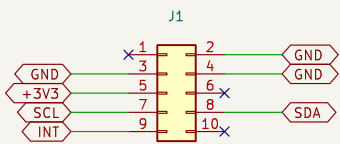


MCU – STM32F030C8T6 running the RetroPad firmware (I2C slave 0x6D)

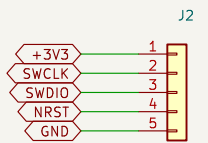


Tab5 Ext.Port1 (M5Stack Tab5 Keyboard P1 pinout) – verify pin 1 orientation before fab



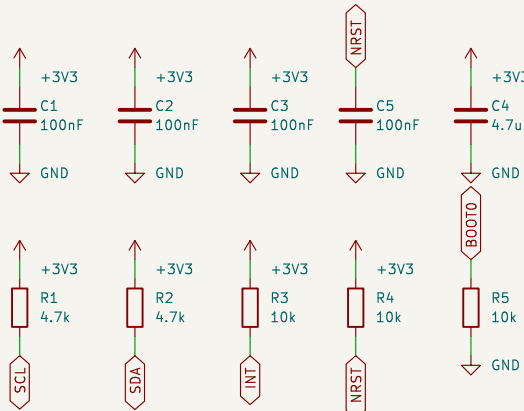
Tab5 Ext.Port1

SWD (same order as the keyboard's P2)

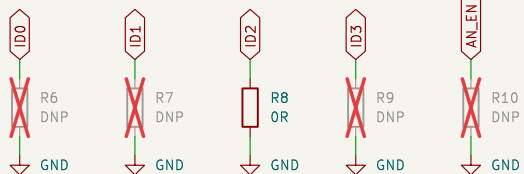


SWD

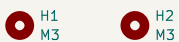
Decoupling, pull-ups, reset, BOOT0 (copied from the M5Stack keyboard)



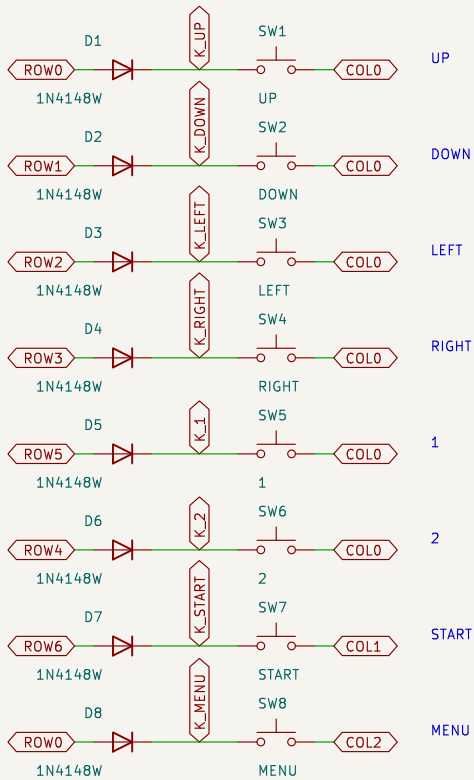
Console ID straps: 0R to GND = bit set. Master System / Game Gear = ID 4 (fit R8; R6, R7, R9, R10 DNP)



Mounting holes (M3, 96 mm apart)



Button matrix – row driven high, column read with pull-down; diode anode on the row



Tab5 keyboard–port board: STM32F030 + RetroPad firmware, I2C 0x6D
Generated by kicad_gen/gen_sch.py from gen_pcb.py and layout.py

RetroESP32–P4

Sheet: /
File: retropad_sms.kicad_sch

Title: RetroPad Master System / Game Gear controller (console ID 4)

Size: A3 Date: 2026–10–05

Rev: 0.1

KiCad E.D.A. kicad-cli 7.0.11+dfsg–1build4

Id: 1/1